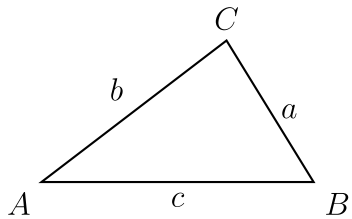
Skills Summary

The Law of Cosines: SAS and SSS Triangles

Use this reference while completing your worksheet or when studying.

The law of sines needs a side and the angle *opposite* it. SAS and SSS never hand you that pair, which is exactly when the law of cosines is the only way in.



How every triangle here is labelled: each side takes the lower-case letter of the vertex opposite it. No values shown — use the numbers given in each problem.

1. SAS: two sides and the angle between them

Formula: $c^2 = a^2 + b^2 - 2ab\cos C$, where C is the angle *between* the two known sides and c is the side opposite it. The letters rotate: for a use $a^2 = b^2 + c^2 - 2bc\cos A$.

Sign check: if C is obtuse, $\cos C$ is negative and the formula *adds*, so the new side is longer than both givens. If $C = 90°$ the term vanishes and it is the Pythagorean theorem.

Example: $a = 6$, $b = 10$, $C = 60°$. Find c.
Step 1. The known angle sits between the two known sides, so there is no side-opposite-angle pair and the law of cosines is the only starting point.
Step 2. $c^2 = 36 + 100 - 2(6)(10)\cos 60° = 136 - 120(0.5) = 76$, and $c = \sqrt{76}$.
$\boldsymbol{c \approx 8.72}$

Try it: $a = 8$, $b = 5$, $C = 75°$. Find c.

check: $\boldsymbol{c \approx 8.26}$

2. SSS: three sides, find an angle

Formula, rearranged for the angle: $\cos C = \dfrac{a^2 + b^2 - c^2}{2ab}$. The side opposite the angle you want is the one that gets *subtracted*; the other two are added and their product is doubled underneath.

Why this beats the law of sines: $\cos^{-1}$ returns angles from $0°$ to $180°$, so an obtuse angle comes out obtuse. A negative cosine means obtuse; a positive one means acute. Start with the largest side to find the largest angle and the rest cannot be obtuse.

Example: $a = 4$, $b = 6$, $c = 8$. Find C.
Step 1. Three sides and no angles, so no ratio of the law of sines can be written; rearrange the law of cosines for the angle opposite c.
Step 2. $\cos C = \dfrac{16 + 36 - 64}{2(4)(6)} = \dfrac{-12}{48} = -0.25$, and the negative value says C is obtuse.
$\boldsymbol{C \approx 104.48°}$

Try it: $a = 7$, $b = 9$, $c = 12$. Find C.

check: $\boldsymbol{C \approx 96.38°}$

3. Bearings and navigation

Turning a story into SAS: draw the two legs from the point they share. The angle at that point is the angle between the courses — for two bearings measured from the same place it is their *difference*, and for a turn along a path it is the interior angle the problem states. Then the distance apart is the third side.

Keep the units of the two legs the same, and give the answer in those units.

Example: Two hikers leave a hut, one walking 3 km and the other 5 km, with 120° between their paths. How far apart are they?
Step 1. The two paths and the line between the hikers form a triangle whose known angle lies between the two known sides — SAS.
Step 2. $c^2 = 9 + 25 - 2(3)(5)\cos 120° = 34 - 30(-0.5) = 49$.
$\boldsymbol{c = 7}$ km

Try it: Two boats leave a jetty, one going 6 km and the other 10 km, with 100° between their courses.

check: $\boldsymbol{c \approx 12.52}$ km